RL Decoders for the Surface Code: A Comparison of DQN and PPO against MWPM

William Seward
Northeastern University
Boston, MA
seward.w@northeastern.edu

Abstract

Decoding the surface code is typically handled by minimum-weight perfect matching (MWPM), which is optimal for independent depolarizing noise but degrades under structured or correlated noise. We compare model-free deep reinforcement learning - Deep Q-Networks (DQN) and Proximal Policy Optimization (PPO) - against an MWPM baseline across four noise models (depolarizing, biased-Z, measurement-dominated, correlated) at code distances $d \in \{3,5\}$. Using a custom Gymnasium MDP backed by the Stim stabilizer simulator, we train both agents for 5×10^5 environment steps and evaluate against PyMatching on identical held-out syndrome sets. At this budget, RL is one to two orders of magnitude worse than MWPM on depolarizing noise; the gap narrows on biased-Z and measurement noise and essentially vanishes on correlated noise, where MWPM's independence assumption is violated. A cross-noise generalization study shows that a DQN trained on depolarizing noise outperforms one trained directly on biased-Z when both are evaluated on biased-Z. Code: https://github.com/sewardwi/qec-rl-comparison.

Introduction

Quantum error correction (QEC) is essential for fault-tolerant quantum computation. Physical qubits are inherently noisy, so logical qubits must be encoded redundantly across many physical qubits, with errors detected via repeated syndrome measurements and corrected by a classical decoder. The surface code is the leading QEC architecture due to its high noise threshold and planar layout compatible with current hardware Fowler et al. (2012). Google recently demonstrated below-threshold surface-code operation on the Willow processor Google Quantum AI (2024), and Google DeepMind's AlphaQubit showed that a neural-network decoder can outperform traditional methods on real experimental data Bausch et al. (2024). The industry challenge is designing decoders that are simultaneously accurate, fast, and adaptive to realistic noise.

The standard classical decoder is minimum-weight perfect matching (MWPM), implemented efficiently by PyMatching Higgott (2022) and used as the baseline in the Willow experiment. MWPM has two drawbacks relevant to this work. First, it requires an explicit, accurate noise model: matching-graph edge weights are negative log-probabilities, so any mismatch between the assumed and actual channel degrades performance. Second, it handles correlations poorly. Real devices exhibit correlated errors from crosstalk, leakage, and coherent single-qubit drift; these violate the independence assumption baked into the matching graph and push MWPM away from the maximum likelihood decoder.

Deep reinforcement learning is an attractive alternative because it sidesteps both drawbacks: an RL decoder has no explicit noise model and instead infers a policy from interaction alone. Several works have applied RL to surface- and toric-code decoding Sweke et al. (2021); Colomer et al. (2020); Fösel et al. (2018); Nautrup et al. (2019), with the broader machine-learning-for-QEC literature surveyed in Wang and Tang (2024). These works generally report that agents can be trained to solve small-distance codes, sometimes approach MWPM performance, and occasionally exceed it on tailored noise. The picture is promising but not settled: environments, reward shapings, and baselines across papers are heterogeneous and direct head-to-head comparisons on identical syndrome sets are rare.

This project contributes a clean comparison of DQN Mnih et al. (2015) and PPO Schulman et al. (2017) against MWPM on the same code instances, the same held-out syndrome sets, and four noise models. The goal is not to beat MWPM on the depolarizing benchmark - not expected at this training scale - but to map out where RL is and is not closing the gap. We release the full stack - a Gymnasium environment backed by Stim Gidney (2021), Stable-Baselines3 Raffin et al. (2021) agents, and a unified evaluation harness using Sinter and PyMatching - as a reusable benchmark.

Our contributions are as follows. (i) We formulate a finite-horizon MDP for surface-code decoding with a tensor observation that encodes both syndrome rounds and accumulated correction state, and an action space that lets the agent commit to any Pauli correction over the data qubits. (ii) We implement and train DQN and PPO agents with a custom small-kernel CNN torso suited to low-dimensional syndrome grids. (iii) We evaluate both agents and the MWPM baseline under four noise models and report logical error rates on identical held-out syndrome sets. (iv) We provide a cross-noise generalization study showing that the generalization behaviour of RL decoders can be non-obvious: training on one noise model sometimes produces lower LER on a different one than training on the evaluation noise directly.

Background

MDPs. A finite-horizon Markov Decision Process is a tuple $(\mathscr{S}, \mathscr{A}, P, R, \gamma, H)$ with state space, action space, transition kernel $P(s' \mid s,a)$, reward function $R(s,a)$, discount factor $\gamma \in (0,1]$, and horizon H Sutton and Barto (2018). A policy

$\pi(a \mid s)$ induces a trajectory distribution and the agent's objective is to maximize $\mathbb{E}_\pi[\sum_{t=0}^{H-1} \gamma^t R(s_t, a_t)]$.

DQN. DQN Mnih et al. (2015) approximates the state-action value function $Q^\pi(s,a) = \mathbb{E}_\pi[\sum_{k=0}^{H-1} \gamma^k r_{t+k} \mid s_t = s, a_t = a]$ with a neural network Q_θ and bootstraps via the Bellman target $y_t = r_t + \gamma \max_{a'} Q_{\theta^-}(s_{t+1}, a')$, where θ^- is a periodically- synced target network. Training minimises $\mathbb{E}_{(s,a,r,s')\sim\mathscr{D}}[(y - Q_\theta(s,a))^2]$ on replay-buffer samples, with ε-greedy exploration.

PPO. PPO Schulman et al. (2017) is an on-policy policy-gradient method that maximises the clipped surrogate objective

$$\mathscr{L}^{\text{CLIP}}(\theta) = \mathbb{E}_t\big[\min\big(\rho_t(\theta)\hat{A}_t, \\ \text{clip}(\rho_t(\theta), 1-\varepsilon, 1+\varepsilon)\hat{A}_t\big)\big] \tag{1}$$

where $\rho_t(\theta) = \pi_\theta(a_t \mid s_t)/\pi_{\theta_{\text{old}}}(a_t \mid s_t)$ is the importance-sampling ratio between updated and behaviour policies and $\hat{A}_t$ is a generalised advantage estimate Schulman et al. (2016). PPO also carries a value-function loss and an entropy bonus and uses mini-batch SGD over several epochs per rollout.

Surface code. The rotated surface code of distance d uses d^2 data qubits arranged on a $d \times d$ lattice, with d^2-1 measurement qubits on the plaquettes. Each plaquette carries either an X-type or a Z-type stabilizer. A round of syndrome extraction measures every stabilizer; a stabilizer flips whenever an odd number of its data qubits suffer anticommuting Pauli errors. The distance d determines how many physical errors are always correctable in the ideal case: $\lfloor (d-1)/2 \rfloor$. A decoder receives the syndrome σ (possibly across d noisy rounds) and returns a Pauli correction C; it succeeds if $E \cdot C$ lies in the stabilizer group and fails otherwise. MWPM solves this by building a graph of syndrome defects, weighting edges by log-probability of the connecting fault path, and returning the minimum-weight perfect matching. This is optimal for independent depolarizing errors but not for correlated noise.

Related Work

Fösel et al. (2018) first demonstrated deep RL for quantum feedback, using an RL agent to discover protective operations against noise on small quantum memories. Although the concrete setting differs from surface-code decoding, the work established that neural RL agents can be trained to produce useful quantum-control policies and introduced many architectural choices - convolutional networks, replay-based training - that later decoders inherited.

Sweke et al. (2021) developed the canonical RL-based surface-code decoder, training a DQN agent with a convolutional Q-network and reporting near-threshold behaviour on distance-three and distance-five codes, along with a Gymnasium-compatible environment that subsequent work builds on. Colomer et al. (2020) apply a similar DQN formulation to the phenomenological toric code. Our work sits most directly alongside these two: we adopt their DQN formulation and extend it by (i) comparing head-to-head with PPO in an identical environment, (ii) using Stim for circuit-level rather than phenomenological noise, and (iii) evaluating on identical held-out syndrome sets against MWPM so comparisons are apples-to- apples. Nautrup et al. (2019) use RL to *discover* QEC codes themselves rather than decode them; that problem is distinct and we do not consider it here.

The most compelling recent learned-decoder result is AlphaQubit Bausch et al. (2024), a transformer-based decoder trained with supervised learning on Google's Willow processor data that outperforms both MWPM and correlated-matching on real experimental syndromes. AlphaQubit demonstrates that neural decoders *can* beat matching given sufficient training data, and clarifies the RL-specific challenge: training RL agents without privileged access to ground-truth corrections is substantially harder than supervised regression. A comprehensive survey of this broader literature is given by Wang and Tang (2024).

Project Description

State. The state is a tensor of shape $(d+2) \times H \times W$, with $H \times W$ matching the syndrome grid of a distance-d rotated surface code. The first d channels are binary masks of triggered stabilizers for each of the d measurement rounds; the last two accumulate the X- and Z-component corrections committed so far during the episode. This gives the policy both the raw evidence of defects and its own running plan, which we found important for PPO in particular. Representative syndrome patterns per noise model are shown in Appendix B.

Actions. The action space $\mathscr{A} = \{0, \ldots, 3d^2\}$ has size $3d^2+1$. Actions 0 through $3d^2-1$ apply one of $\{X, Y, Z\}$ to a specific data qubit, updating the correction accumulators but not consuming the round; action $3d^2$ commits the accumulated correction and advances the episode, matching the "request next syndrome" action family in Sweke et al. (2021). Committing is not free: after d commits the environment applies the full accumulated correction to the simulated state, measures the logical operator, and emits a terminal reward. A worked decoder episode is shown in Appendix C.

Reward. Pure sparse rewards on logical success $(+1/-1)$ produced unusably weak gradients. We use the combined shaping

$$r_t = \alpha\, \Delta w_t - \beta\, 1[a_t \neq \text{commit}] + r_t^{\text{term}}, \tag{2}$$

where $\Delta w_t = w(\sigma_{t-1}) - w(\sigma_t)$ is the reduction in syndrome weight induced by the agent's Pauli action - a potential-based term that does not change the optimal policy Ng (Harada) - the second term is a small per-action penalty to discourage runaway corrections, and $r_t^{\text{term}} \in \{-1, 0, +1\}$ is a sparse terminal reward on logical success $(+1)$, identity correction (0), or logical error (-1). During *evaluation* we always use the pure sparse terminal reward regardless of the reward used during training, so comparisons are fair.

Transitions and discount. Transitions are deterministic given the agent's actions: applying a Pauli to a data qubit flips the adjacent stabilizers and updates the correction channels; committing advances to the next round. Stochasticity comes from noise injection at episode reset, where we draw one fresh syndrome instance per episode from Stim. We use $\gamma = 0.99$.

Agents and architecture. Both agents share a custom convolutional torso, `SyndromeCNN`: three 3×3 stride-1 convolutions with ReLU followed by a fully connected projection to 256 features. The standard Nature-DQN torso (8×8 stride 4) collapses the small syndrome grids on its first layer; the small-kernel design follows prior RL-QEC practice Sweke et al. (2021); Colomer et al. (2020). DQN attaches a dueling head producing Q-values over the action space; PPO attaches separate policy and value heads. DQN uses a 10^5-transition replay buffer, target-network updates every 10^3 steps, Adam at 5×10^{-4}, and ε annealed linearly from 1.0 to 0.05 over the first 10% of training. PPO uses a 2048-step rollout, minibatches of 64, four epochs per rollout, Adam at 3×10^{-4}, and clip $\varepsilon = 0.2$. Both come from Stable-Baselines3 Raffin et al. (2021) with the torso replaced.

Noise models. We implement four noise models in Stim (Table 1). Depolarizing: each operation is followed by an X, Y, or Z error at rate $p/3$. Biased-Z: Z probability $0.9p$ and X, Y probability $0.05p$ each, reflecting the dephasing-dominated regime of superconducting transmons. Measurement: readout error inflated to $10p$ with gate errors at p. Correlated: two-qubit error terms that flip adjacent data qubits jointly with a small probability, modelling crosstalk.

Model	Physical motivation
Depolarizing	Symmetric $X/Y/Z$, canonical benchmark
Biased-Z	Dephasing-dominated, as in transmons
Measurement	Readout errors 10× gate errors
Correlated	Crosstalk between neighboring qubits

Table 1: Noise models. Correlated noise violates MWPM's independence assumption.

Training and evaluation. Syndromes are pre-sampled in batches of 1024 for efficient environment resets; evaluation uses deterministic held-out syndrome sets of 5000 shots each to remove variance from the cross-decoder comparison. Every agent trains for 5×10^5 environment steps. Evaluation uses 500 episodes, and MWPM is evaluated on the exact same held-out syndromes so logical error rates (LER) are directly comparable: LER $=$ #{episodes with logical error}/#{episodes}.

Experiments

The proposal originally targeted $d \in \{3, 5, 7\}$ for all three decoders. Training the RL agents at $d = 7$ proved intractable at our compute budget - the action space grows from 76 at $d = 5$ to 148 at $d = 7$, with total detector-error-model mechanisms growing roughly as d^3 (Appendix A) - and the RL sweep was restricted to $d \in \{3, 5\}$ as anticipated in the proposal's risk section. MWPM was extended to $d = 7$ as a baseline validation (see Figure 7 below).

Training Dynamics

Figure 1 shows training curves for DQN and PPO at $d = 3$ on depolarizing noise. Both agents achieve comparable mean shaping reward (0.5–0.65) within the first 100k environment steps. DQN exhibits the characteristic instability of value-based methods at this scale: a collapse in mean reward from $+0.65$ to -0.5 near step 100k with a corresponding LER spike to 0.78, followed by recovery; a second smaller LER spike near step 275k; and a final instability in the last 50k steps that reopens the LER back toward 0.47. PPO is visibly smoother but not monotonic, and neither curve is clearly descending by cutoff. Reading these curves charitably, the practical converged LER at $d = 3$ is around 0.2 for both agents - an order of magnitude above MWPM's LER in the same regime.

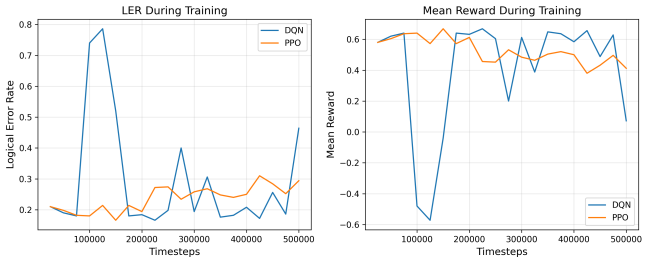

Figure 1: Training curves at $d = 3$ on depolarizing noise. Left: evaluation LER. Right: mean shaping reward. DQN is noticeably less stable than PPO.

The instability in DQN's training reflects the usual value-function drift problem: when the target network has not yet caught up to recent policy improvements, $\max_{a'} Q_{\theta^-}$ can be systematically biased, and for this environment that bias manifests as episodes in which the agent commits corrections that make things worse rather than better. PPO's smoother training curve is consistent with the general trend that on-policy clipped methods are less prone to this failure mode at modest compute budgets.

Primary Comparison

Figure 2 shows the evaluation LER of DQN, PPO, and MWPM under all four noise models at $d = 3$ and $d = 5$.

On the canonical depolarizing benchmark MWPM achieves LER $\sim 10^{-3}$ at $p = 10^{-3}$ and scales as expected with p. At the same physical error rate the RL agents are at LER $\sim 5 \times 10^{-2}$ (DQN) and $\sim 10^{-1}$ (PPO), one to two orders of magnitude above MWPM. The expected code-distance benefit of going from $d = 3$ to $d = 5$ is visible for MWPM but is inverted or absent for the RL agents, whose $d = 5$ curves are typically at or above their $d = 3$ curves. This is strong evidence that the RL agents are under-trained at $d = 5$: the action space grows from 28 to 76 entries and the training budget has not scaled with it.

On measurement-dominated noise the picture is similar. MWPM again dominates at small p and shows the expected

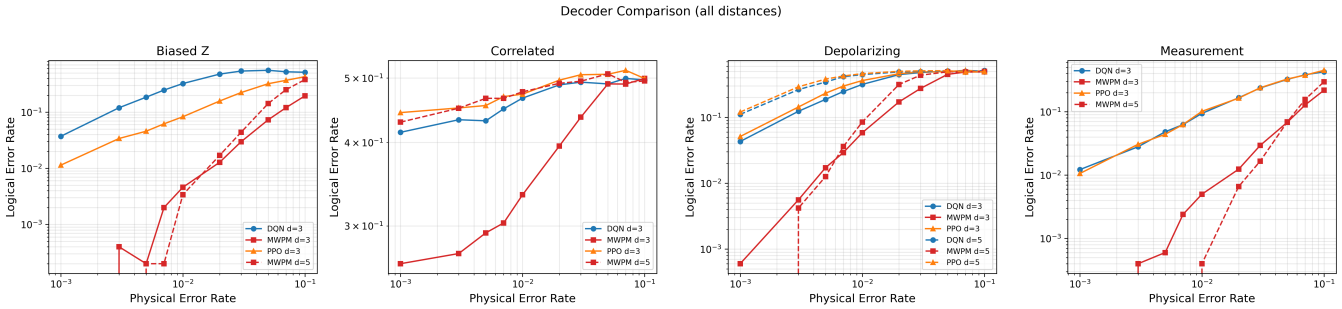

Figure 2: LER vs. physical error rate for DQN, PPO, and MWPM across all four noise models at $d = 3$ (solid) and $d = 5$ (dashed). MWPM dominates on depolarizing and measurement noise; on correlated noise all decoders saturate near the random baseline; on biased-Z RL reaches intermediate performance. The expected code-distance improvement for MWPM is recovered on depolarizing and measurement noise but not for RL.

d-scaling. DQN and PPO are comparable to each other and substantially worse than MWPM, though the gap is somewhat smaller than on pure depolarizing noise.

On biased-Z noise the overall LER levels are lower for everyone and the gap between RL and MWPM is narrower at small p than on depolarizing noise. PPO outperforms DQN on this noise model, especially at the smallest p - a clean case in which the on-policy method's smoother learning translates to lower LER at comparable compute.

On correlated noise the story is qualitatively different. All decoders - including MWPM - have LER in the 0.3–0.5 range across the entire p sweep; none is meaningfully below the others. This is not a success for RL so much as a failure of everything: the correlated-crosstalk channel at these rates is essentially undecodable at $d = 3$. That said, the *relative* gap between RL and MWPM is narrowest here, because MWPM's matching graph is mis-calibrated for correlations and cannot exploit them.

Figure 3 quantifies the gap: the LER ratio of each RL decoder to MWPM on identical held-out syndrome sets. A value of one indicates parity with MWPM. Ratios are largest at small p (where MWPM's denominator LER shrinks faster than the RL agents') and converge toward one at $p \sim 10^{-1}$ where all decoders saturate. The largest isolated ratio, $\sim 900\times$, is a DQN data point near $p = 5 \times 10^{-3}$; the median ratio across noise models is 50–100$\times$ at small p and 2–5$\times$ at $p \geq 3 \times 10^{-2}$.

Code-Distance Scaling

Figure 4 decomposes the full p-sweep per-decoder at $d \in \{3, 5\}$. The MWPM panel shows the textbook crossing between distances: below threshold, larger d suppresses logical errors exponentially; above threshold, larger d makes things worse. A dedicated MWPM-only threshold study at $d \in \{3, 5, 7\}$ (Appendix B) reproduces $p_{\text{th}} \approx 7 \times 10^{-3}$, consistent with the surface-code literature Fowler et al. (2012) and validating the Stim circuit, the PyMatching decoder, and our evaluation pipeline. The DQN and PPO panels, by contrast, show flat curves with $d = 5$ typically at or above $d = 3$. This is quantitative evidence that the RL agents at $d = 5$ are not yet in a fault-tolerant regime at our training budget; recovering code-distance scaling for RL decoders would require substantially longer training or a curriculum over d, as in Sweke et al. (2021).

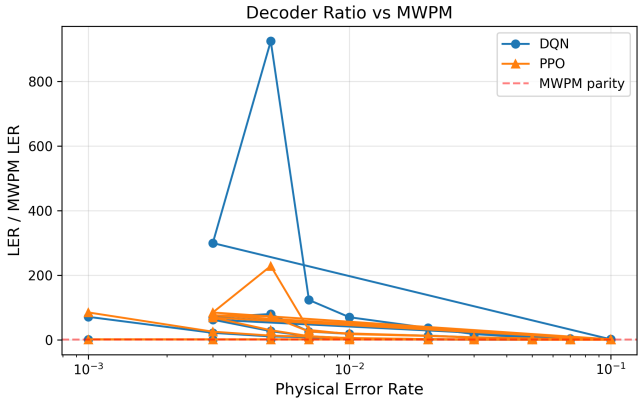

Figure 3: LER ratio of each RL decoder to MWPM on identical held-out syndrome sets. A value of one indicates parity with MWPM; curves converge toward one at high p where every decoder is at the random baseline.

Cross-Noise Generalization

We trained a DQN and a PPO agent on each of two noise models (biased-Z and depolarizing) and then evaluated every trained agent on all four noise models. Figure 5 shows the resulting 2×4 matrices per algorithm. Three observations stand out.

First, correlated noise is hard regardless of training noise: every column labelled *correlated* shows the worst LER in its row, for both algorithms. This is consistent with the primary-comparison result that the correlated-crosstalk regime is genuinely difficult.

Second, measurement-dominated noise is the opposite: every *measurement* column shows the best or near-best LER in its row, regardless of which noise the agent was trained on. The measurement model's noise structure (many, rare, temporally extended defects) appears to be generically easy for the learned CNN features.

Third, and most interestingly, training on one noise model can outperform training directly on the evaluation noise. DQN trained on depolarizing noise achieves LER 0.042 on biased-Z evaluation - *better* than DQN trained on biased-Z

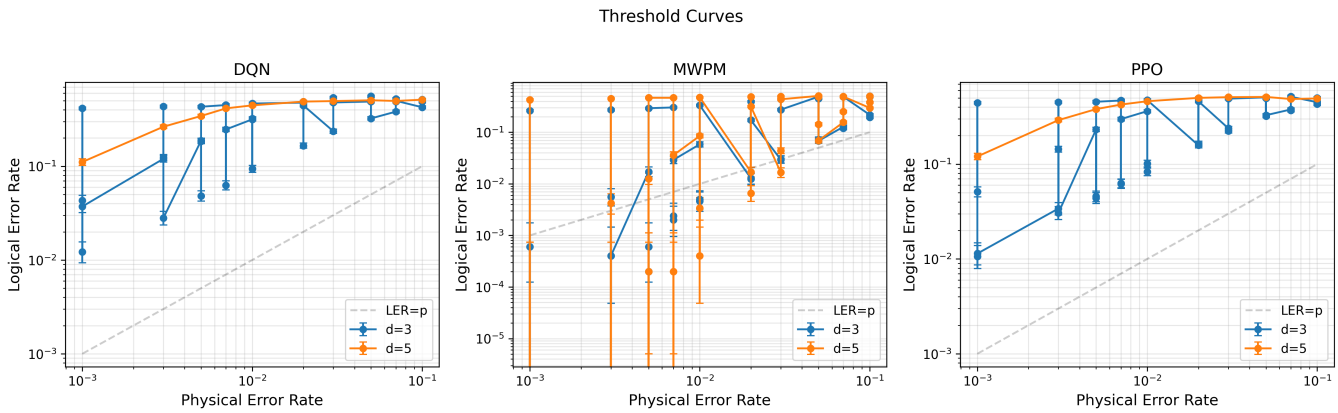

Figure 4: Per-decoder threshold curves at $d \in \{3,5\}$. MWPM (center) displays the expected crossing between distances; DQN (left) and PPO (right) show no such crossing and in fact have $d = 5$ above $d = 3$, indicating that the $d = 5$ agents are under-trained at 5×10^5 steps.

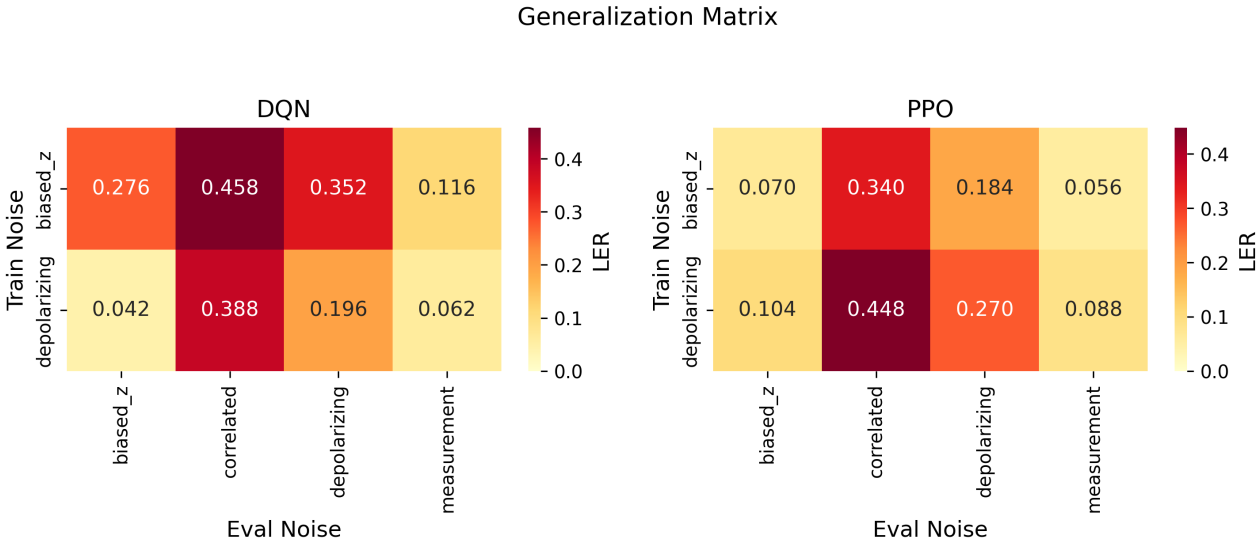

Figure 5: Cross-noise generalization matrices. Rows are training noise, columns are evaluation noise; cells are LER on the held-out 500-episode set at $p = 10^{-2}$. Lower is better. The best DQN cell for biased-Z evaluation (0.042) is obtained by the *depolarizing*-trained agent, not the biased-Z-trained one.

itself (0.276) - and likewise achieves LER 0.062 on measurement evaluation, better than any other configuration. The most plausible explanation is that depolarizing noise is a *harder* training distribution than biased-Z: it exposes the agent to X, Y, and Z defects in roughly equal proportion and forces a more general representation, which transfers favourably to the easier biased-Z and measurement regimes. This is consistent with findings in the broader neural-decoder literature Bausch et al. (2024); Wang and Tang (2024).

Discussion and Limitations

Three observations are worth highlighting.

First, the canonical RL-vs-MWPM gap is real and large. On depolarizing noise at our training budget, RL is one to two orders of magnitude worse than MWPM, with the largest ratios appearing precisely where MWPM is most confident. This is a direct empirical contradiction of any narrative in which half-a-million- step RL agents are plausibly competitive on the canonical benchmark. The existing path to a neural decoder that does beat MWPM is supervised, not reinforcement-learning-based, as demonstrated by AlphaQubit Bausch et al. (2024); closing this gap with RL remains open.

Second, the RL agents do not recover code-distance scaling. Going from $d = 3$ to $d = 5$ increases the action space from 28 to 76 while holding the training budget fixed, and the resulting $d = 5$ agents are effectively untrained. Any RL-for-QEC argument that wants to claim fault-tolerance scaling needs to address this; the most promising directions in the literature involve curricula in p and d Sweke et al. (2021).

Third, the interesting regimes for RL are the ones where MWPM itself is weak. Correlated noise is the cleanest example: it is a noise model on which matching's inductive bias is wrong. Our results on this model show all decoders at simi-

lar poor LER, but the gap to MWPM is narrowest here, and the cross-noise generalization study shows that an RL agent's representation can transfer in non-obvious ways across noise models. This is the natural follow-up research direction: dedicate compute to the noise models for which matching is not the right baseline.

Our experiments are limited by compute. We train for 5×10^5 steps per agent; published works that narrow the gap to MWPM generally train for 10^7 steps or more and with curriculum schedules Sweke et al. (2021). We evaluate at $d \in \{3, 5\}$ only, as anticipated in the proposal's risk analysis; fault-tolerant regimes of interest are at $d \geq 7$. Our correlated noise model is a stylised crosstalk channel, not a calibration-fit device model. A full-budget reward-shaping ablation (see Appendix D for a short-horizon version) is also future work.

Conclusion

We compared model-free deep RL decoders (DQN and PPO) against classical MWPM for the distance-three and distance-five surface code under four noise models. At a training budget of 5×10^5 environment steps, neither RL agent matches MWPM on depolarizing noise: the gap is typically one-to-two orders of magnitude in LER, with the largest ratios at small p where MWPM is most confident. The gap narrows substantially on structured noise (biased-Z, measurement-dominated) and essentially vanishes on correlated-crosstalk noise, where MWPM's independence assumption is violated and all decoders saturate near the random baseline. A cross-noise generalization study shows that an RL decoder's representation can transfer in non-obvious ways: a DQN trained on depolarizing noise can outperform one trained directly on biased-Z when both are evaluated on biased-Z. These observations are consistent with the industry picture of neural decoders set by Google's Willow experiment Google Quantum AI (2024) and AlphaQubit Bausch et al. (2024): machine-learning-based decoders are a real path to beating matching, but doing so with online reinforcement learning - rather than supervised learning on logged syndromes - is substantially harder. Natural next steps are longer training, larger code distance, curriculum over p and d, and realistic noise channels derived from device calibration data. All code, configurations, and evaluation sweeps are available at https://github.com/sewardwi/qec-rl-comparison.

References

Bausch, J.; Senior, A. W.; Heras, F. J. H.; et al. 2024. Learning high-accuracy error decoding for quantum processors. *Nature*. https://www.nature.com/articles/s41586-024-08148-8.

Colomer, L. D.; Skotiniotis, M.; and Muñoz-Tapia, R. 2020. Reinforcement learning for optimal error correction of toric codes. *Physics Letters A*, 384(17): 126353.

Fösel, T.; Tighineanu, P.; Weiss, T.; and Marquardt, F. 2018. Reinforcement learning with neural networks for quantum feedback. *Physical Review X*, 8(3): 031084.

Fowler, A. G.; Mariantoni, M.; Martinis, J. M.; and Cleland, A. N. 2012. Surface codes: Towards practical large-scale quantum computation. *Physical Review A*, 86: 032324.

Gidney, C. 2021. Stim: a fast stabilizer circuit simulator. *Quantum*, 5: 497.

Google Quantum AI. 2024. Quantum error correction below the surface code threshold. *Nature*. https://www.nature.com/articles/s41586-024-08449-y.

Higgott, O. PyMatching: A Python/C++ library for decoding quantum codes with minimum-weight perfect matching. https://github.com/oscarhiggott/PyMatching.

Mnih, V.; Kavukcuoglu, K.; Silver, D.; Rusu, A. A.; Veness, J.; Bellemare, M. G.; Graves, A.; Riedmiller, M.; Fidjeland, A. K.; Ostrovski, G.; et al. 2015. Human-level control through deep reinforcement learning. *Nature*, 518(7540): 529–533.

Nautrup, H. P.; Delfosse, N.; Dunjko, V.; Briegel, H. J.; and Friis, N. 2019. Optimizing quantum error correction codes with reinforcement learning. *Quantum*, 3: 215.

Ng, A. Y.; Harada, D.; and Russell, S. J. 1999. Policy invariance under reward transformations: Theory and application to reward shaping. In *Proceedings of the International Conference on Machine Learning*, 278–287.

Raffin, A.; Hill, A.; Gleave, A.; Kanervisto, A.; Ernestus, M.; and Dormann, N. 2021. Stable-Baselines3: Reliable reinforcement learning implementations. *Journal of Machine Learning Research*, 22(268): 1–8.

Schulman, J.; Moritz, P.; Levine, S.; Jordan, M. I.; and Abbeel, P. 2016. High-dimensional continuous control using generalized advantage estimation. In *International Conference on Learning Representations*.

Schulman, J.; Wolski, F.; Dhariwal, P.; Radford, A.; and Klimov, O. 2017. Proximal policy optimization algorithms. *arXiv preprint* arXiv:1707.06347.

Sutton, R. S.; and Barto, A. G. 2018. *Reinforcement Learning: An Introduction*. MIT Press, 2nd edition.

Sweke, R.; Kesselring, M. S.; van Nieuwenburg, E. P. L.; and Eisert, J. 2021. Reinforcement learning decoders for fault-tolerant quantum computation. *Machine Learning: Science and Technology*, 2(2): 025005.

Wang, Z.; and Tang, H. 2024. Artificial intelligence for quantum error correction: A comprehensive review. *arXiv preprint* arXiv:2412.20380.

Technical Appendix

The following supplementary figures support the main paper.

A. Circuit Complexity Scaling

Figure 6 gives the dimensions of the Stim-generated circuits and the induced RL environment at each code distance. $|\mathscr{A}| = 3d^2 + 1$ is the action-space cardinality; DEM errors is the number of independent error mechanisms in Stim's detector error model; Total Gates is the depth of one syndrome-extraction round. The DEM-error column grows roughly as d^3, which accounts for the sharp increase in training difficulty from $d = 3$ to $d = 5$ to $d = 7$ and motivates the decision to restrict the RL sweep to $d \in \{3, 5\}$.

Circuit Complexity Scaling

d	Data Qubits	Total Qubits	Detectors	|A|	DEM Errors	Total Gates
3	9	26	24	28	286	54
5	25	64	120	76	1953	86
7	49	118	336	148	6602	118

Figure 6: Circuit and MDP scaling with code distance.

B. MWPM Baseline Threshold Validation

Figure 7 runs MWPM alone at $d \in \{3, 5, 7\}$ on depolarizing noise. The curves cross near $p \approx 7 \times 10^{-3}$, reproducing the expected surface-code threshold behaviour Fowler et al. (2012): below threshold, larger d suppresses the logical error rate exponentially; above threshold, larger d makes things worse. This sanity-checks the Stim circuit construction, the PyMatching decoder, and our evaluation pipeline.

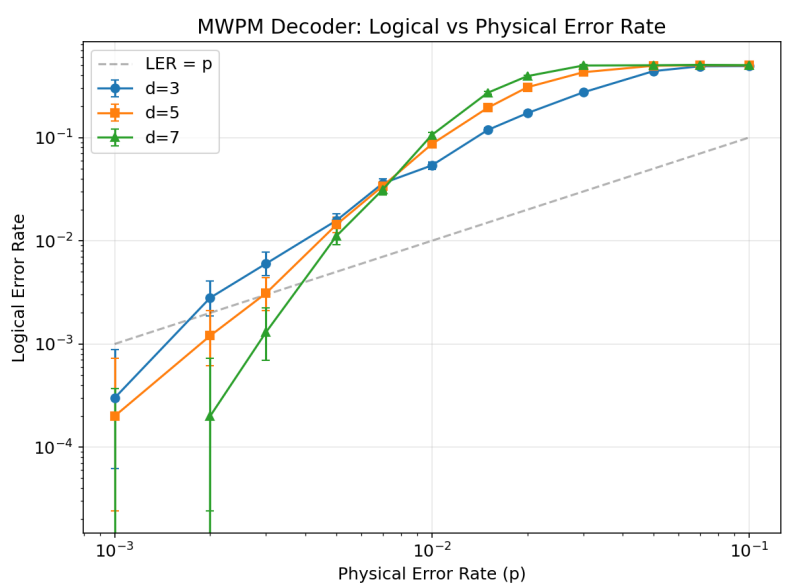

Figure 7: MWPM baseline threshold curves on depolarizing noise at $d \in \{3, 5, 7\}$. Curves cross around $p \approx 7 \times 10^{-3}$, consistent with the literature surface-code threshold.

C. Representative Syndrome Patterns

Figure 8 shows representative $d = 3$ syndrome patterns (round 0) for each noise model at a physical error rate of $p = 10^{-2}$. Marked cells indicate triggered stabilizers. The spatial distribution of defects differs by noise model: depolarizing noise tends to produce isolated defects, while correlated noise produces adjacent pairs.

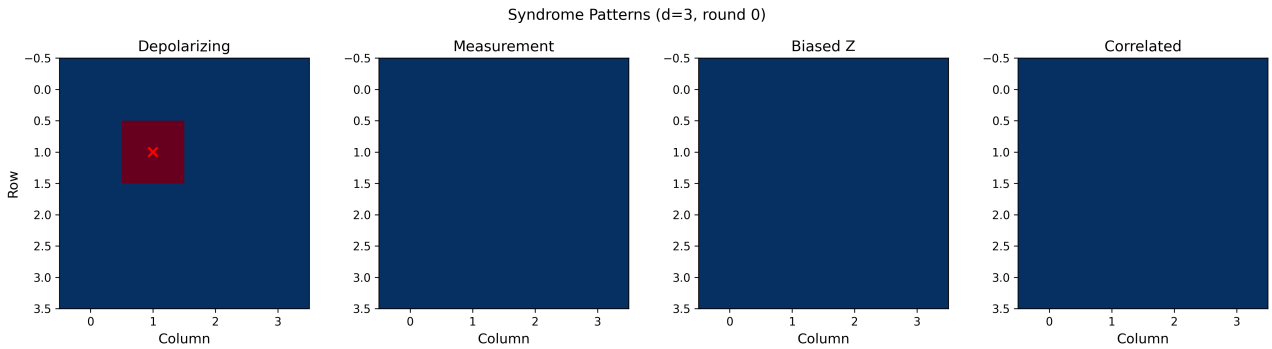

Figure 8: Representative $d = 3$ syndrome patterns, round 0, at $p = 10^{-2}$.

D. Decoder Behavior on a Single Episode

Figure 9 shows a single successful decoding episode at $d = 3$. Left: the syndrome grid observed by the agent at round 0. Right: the accumulated correction at commit time, with X corrections in red and Z corrections in blue. The trial returns logical success.

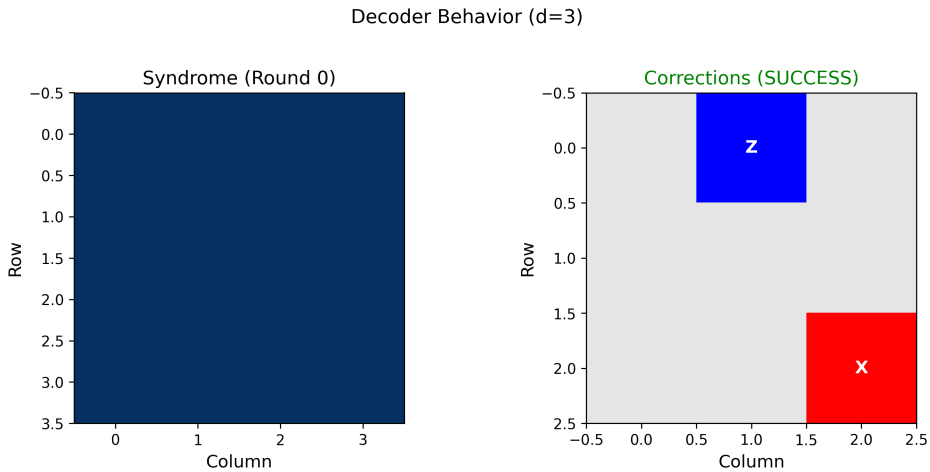

Figure 9: A single successful decoding episode at $d = 3$.

E. Short-Horizon Reward-Shaping Ablation

Figure 10 compares three reward schemes over the first 10k steps of training: the pure sparse terminal reward (the $+1/-1$ form proposed in the project proposal), the potential-based shaping described in the main text, and an additional heuristic-shaped variant that adds a small positive bonus for every stabilizer resolved. At this short training horizon the three schemes produce very similar LER trajectories (~ 0.16–0.27) and similar mean reward; the potential-based scheme shows the most variance. We thus cannot claim from this evidence that shaping is essential at 10k steps. At our full training budget (5×10^5 steps) the potential-based scheme was used throughout and a full-budget ablation was not run; a proper ablation is future work.

F. Comprehensive Noise-Model Overlay

Figure 11 overlays every decoder on every noise model in a single pair of panels (one per code distance). This is redundant with Figure 2 in the main body but is sometimes easier to read for a specific decoder-noise pair because all curves share a single axis. The $d = 5$ panel contains fewer curves because several RL runs at $d = 5$ were omitted from the sweep for compute reasons.

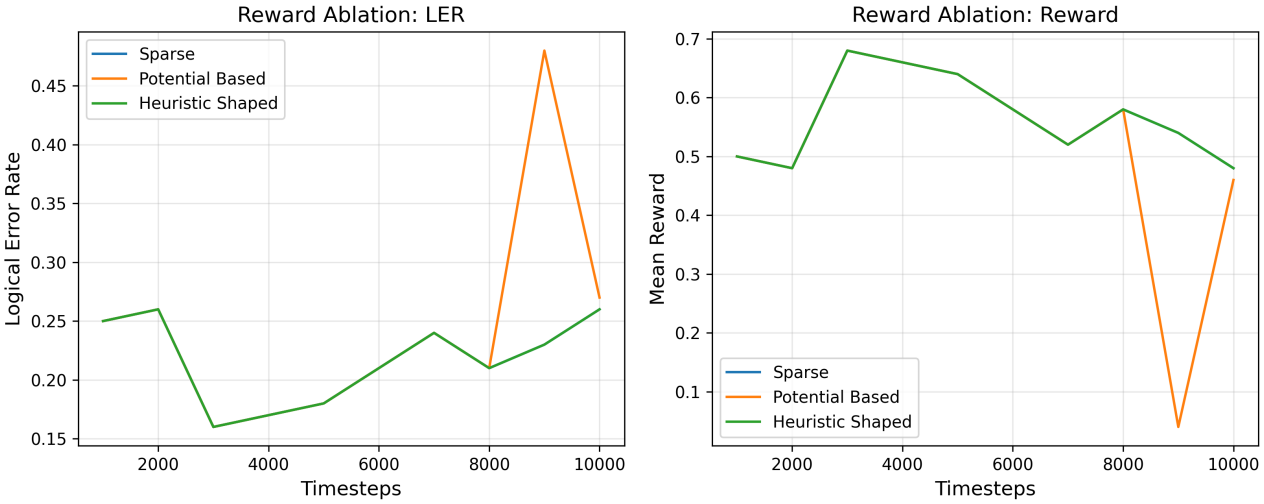

Figure 10: Reward-shaping ablation over the first 10k training steps at $d = 3$ on depolarizing noise.

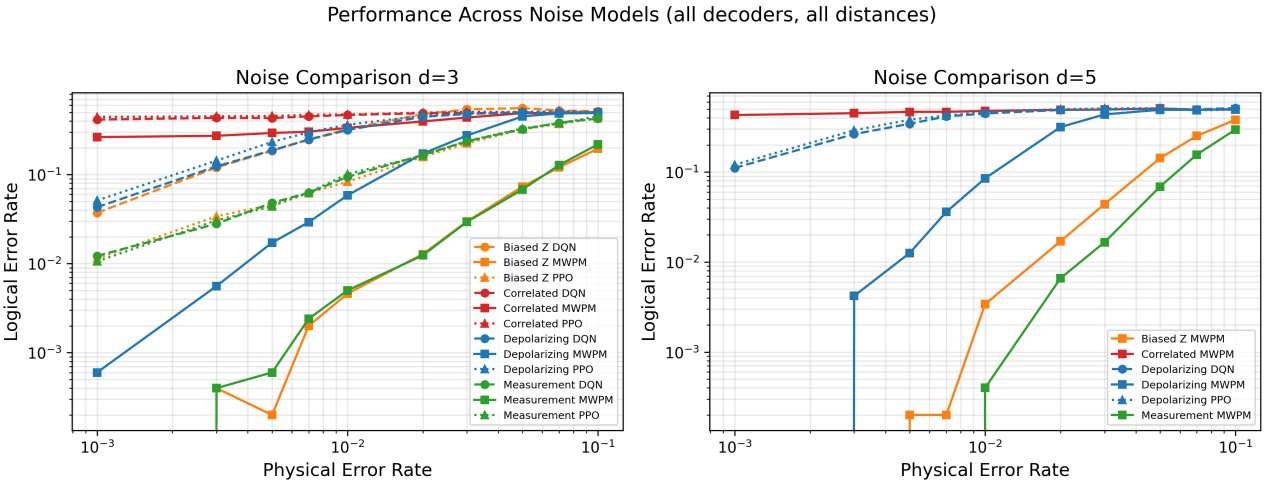

Figure 11: Comprehensive LER vs. p overlay across decoders and noise models at $d = 3$ (left) and $d = 5$ (right). Same data as Figure 2, alternative layout.